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Progress CaseMethod 2

- Class Pembeli29

```
package CaseMethod2;

public class Pembeli29 {
    int noAntrian;
    String pembeli;
    String nohp;

    public Pembeli29(int noAntrian, String pembeli, String nohp){
        this.noAntrian = noAntrian;
        this.pembeli = pembeli;
        this.nohp = nohp;
    }

    public void tampil() {
        System.out.println(noAntrian + "\t" + pembeli + "\t" + nohp);
    }
}
```

- Class Pesanan29

```
package CaseMethod2;

public class Pesanan29 {
    int kodePesanan;
    String namaPesanan;
    int harga;

    public Pesanan29(int kodePesanan, String namaPesanan, int harga) {
        this.kodePesanan = kodePesanan;
        this.namaPesanan = namaPesanan;
        this.harga = harga;

    }
    public void tampil() {
        System.out.println(kodePesanan + "\t" + namaPesanan + "\t" + harga);
    }
    public String getNamaPesanan() {
        return namaPesanan;
    }
}
```

- Class NodePembeli29

```
package CaseMethod2;

public class NodePembeli29 {
    Pembeli29 data;
    NodePembeli29 prev;
    NodePembeli29 next;

    public NodePembeli29(Pembeli29 data) {
        this.data = data;
        this.prev = null;
        this.next = null;
    }
}
```

- Class NodePesanan29

```
package CaseMethod2;

public class NodePesanan29 {
    Pesanan29 data;
    String namaPembeli;
    int noAntrian;
    NodePesanan29 prev;
    NodePesanan29 next;

    public NodePesanan29(Pesanan29 data, String namaPembeli, int noAntrian)
{
        this.data = data;
        this.namaPembeli = namaPembeli;
        this.noAntrian = noAntrian;
        this.prev = null;
        this.next = null;
    }
}
```

- Class DoubleLinkedListPembeli29

```
package CaseMethod2;

public class DoubleLinkedListPembeli29 {
    NodePembeli29 head;
    NodePembeli29 tail;

    public DoubleLinkedListPembeli29() {
        head = null;
        tail = null;
    }

    public boolean isEmpty() {
        return head == null;
    }

    public void addLast(Pembeli29 data) {
        NodePembeli29 newNode = new NodePembeli29(data);
```

```
        if (isEmpty()) {
            head = tail = newNode;
        } else {
            tail.next = newNode;
            newNode.prev = tail;
            tail = newNode;
        }
        System.out.println("Antrian berhasil ditambahkan dengan nomor: " +
data.noAntrian);
    }

    public Pembeli29 removefirst(int noAntrian) {
        if (isEmpty()) {
            System.out.println("Linked List kosong!");
            return null;
        }

        NodePembeli29 current = head;
        while (current != null && current.data.noAntrian != noAntrian) {
            current = current.next;
        }

        if (current == null) {
            System.out.println("Data dengan No Antrian " + noAntrian + " tidak
ditemukan.");
            return null;
        }

        Pembeli29 dataTerhapus = current.data;

        if (head == tail) {
            head = tail = null;
        } else if (current == head) {
            head = head.next;
            head.prev = null;
        } else if (current == tail) {
            tail = tail.prev;
            tail.next = null;
        } else {
            current.prev.next = current.next;
            current.next.prev = current.prev;
        }

        return dataTerhapus;
    }

    public void print() {
        if (isEmpty()) {
            System.out.println("Linked List masih kosong.");
            return;
        }
        System.out.println("Daftar Antrian Pembeli");
        System.out.println("==============================");
```

```
        System.out.println("No Antrian\tNama\tNo HP");
        NodePembeli29 current = head;
        while (current != null) {
            current.data.tampil();
            current = current.next;
        }
        System.out.println("==============================");
    }
}
```

- Class DoubleLinkedListPesanan29

```
package CaseMethod2;

public class DoubleLinkedListPesanan29 {
    NodePesanan29 head;
    NodePesanan29 tail;
    int totalPendapatan = 0;

    public DoubleLinkedListPesanan29() {
        head = null;
        tail = null;
    }

    public boolean isEmpty() {
        return head == null;
    }

    public void addLast(Pesanan29 data, String namaPembeli, int noAntrian) {
        NodePesanan29 newNode = new NodePesanan29(data, namaPembeli,
noAntrian);
        if (isEmpty()) {
            head = tail = newNode;
        } else {
            tail.next = newNode;
            newNode.prev = tail;
            tail = newNode;
        }
        totalPendapatan += data.harga;
        System.out.println(namaPembeli + " telah memesan " +
data.namaPesanan);
    }

    public void sortByNamaPesanan() {
        if (isEmpty() || head == tail)
            return;
        boolean swapped;
        do {
            swapped = false;
            NodePesanan29 current = head;

            while (current != null && current.next != null) {
```

```
                if
(current.data.namaPesanan.compareToIgnoreCase(current.next.data.namaPesa
nan) > 0) {
                    Pesanan29 tempData = current.data;
                    String tempNama = current.namaPembeli;
                    int tempNo = current.noAntrian;

                    current.data = current.next.data;
                    current.namaPembeli = current.next.namaPembeli;
                    current.noAntrian = current.next.noAntrian;

                    current.next.data = tempData;
                    current.next.namaPembeli = tempNama;
                    current.next.noAntrian = tempNo;

                    swapped = true;
                }
                current = current.next;
            }
        } while (swapped);
    }

    public void print() {
        if (isEmpty()) {
            System.out.println("Belum ada pesanan yang masuk.");
            return;
        }

        sortByNamaPesanan();

        System.out.println("Laporan Pesanan");
        System.out.println("================================");
        System.out.println("Kode Pesanan\tNama Pesanan\tHarga");
        NodePesanan29 current = head;
        while (current != null) {
            current.data.tampil();
            current = current.next;
        }
        System.out.println("================================");
        System.out.println("TOTAL PENDAPATAN: Rp " + totalPendapatan);
    }
}
```

- Class MainMenu29

```
package CaseMethod2;

import java.util.Scanner;

public class MainMenu29 {
    static DoubleLinkedListPembeli29 antrian = new
DoubleLinkedListPembeli29();
    static DoubleLinkedListPesanan29 laporan = new
DoubleLinkedListPesanan29();
```

```
    static int nextNoAntrian = 1;

    public static void main(String[] args) {
        Scanner scan = new Scanner(System.in);
        int pilihan;

        // antrian.addLast(new Pembeli29(nextNoAntrian++, "Ainara",
"08224500000"));
        // antrian.addLast(new Pembeli29(nextNoAntrian++, "Danra",
"08224511111"));
        // antrian.addLast(new Pembeli29(nextNoAntrian++, "Sanri",
"08224522222"));

        do {
            System.out.println("\nSISTEM ANTRIAN ROYAL DELISH");
            System.out.println("============================");
            System.out.println("1. Tambah Antrian");
            System.out.println("2. Cetak Antrian");
            System.out.println("3. Hapus Antrian dan Pesan");
            System.out.println("4. Laporan Pesanan");
            System.out.println("0. Keluar");
            System.out.print("Pilih menu : ");
            pilihan = scan.nextInt();
            scan.nextLine();

            switch (pilihan) {
                case 1:
                    System.out.print("Nama Pembeli : ");
                    String nama = scan.nextLine();
                    System.out.print("No HP : ");
                    String noHp = scan.nextLine();
                    Pembeli29 pembeliBaru = new Pembeli29(nextNoAntrian, nama,
noHp);
                    antrian.addLast(pembeliBaru);
                    nextNoAntrian++;
                    break;

                case 2:
                    antrian.print();
                    break;

                case 3:
                    if (antrian.isEmpty()) {
                        System.out.println("Antrian kosong!");
                        break;
                    }
                    antrian.print();
                    System.out.print("Masukkan No Antrian yang dipanggil : ");
                    int no = scan.nextInt();
                    scan.nextLine();

                    Pembeli29 p = antrian.removefirst(no);
                    if (p != null) {
```

```
                    System.out.print("Kode Pesanan : ");
                    int kode = scan.nextInt();
                    scan.nextLine();
                    System.out.print("Nama Pesanan : ");
                    String namaPesanan = scan.nextLine();
                    System.out.print("Harga : ");
                    int harga = scan.nextInt();
                    scan.nextLine();

                    Pesanan29 pesanan = new Pesanan29(kode, namaPesanan,
harga);
                    laporan.addLast(pesanan, p.pembeli, no);
                }
                break;

            case 4:
                laporan.print();
                break;

            case 0:
                System.out.println("Program selesai.");
                break;

            default:
                System.out.println("Menu tidak valid.");
          }
        } while (pilihan != 0);

        scan.close();
    }
}
```